TOMA SASAKI

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EDUCATION

The University of Manchester *Sep 2026 – Present*
BSc Computer Science and Mathematics

INTO Manchester *Sep 2025 – Jun 2026*
International Foundation in Mathematics and Computer Science
Grades: Technical Maths (A*), Physics (A*), Chemistry (A*), English for Academic Purposes (A)

Sendai Johnan High School, Sendai, Japan *Apr 2022 – Mar 2025*
High School Diploma

PROJECTS

Type-safe GUI DSL in Haskell
- Designed a typed DSL for describing GUI component hierarchies, using type class constraints to enforce valid parent-child relationships at compile time.
- Encoded component-specific property constraints in the type system to prevent invalid property assignments.
- Implemented an interpreter to translate DSL descriptions into sequences of rendering commands.

Bare-Metal AVR Breakout Game
- Developed a Breakout-style game in C for an ATmega328P microcontroller using bare-metal programming.
- Designed a custom PCB in KiCad, manufactured through JLCPCB and assembled and tested the hardware.
- Implemented 8x8 LED matrix multiplexing, timer and external interrupt handling and button input.

Native Serial File Transfer Application
- Developed a native Windows GUI application in C using the Win32 API to transfer files over serial connections.
- Tested on Windows 2000 through Windows 11 without requiring additional DLL dependencies.

Typed CLI Input Library in Haskell
- Designed a typed API for interactive command-line input, supporting extensible parsing and validation through type class instances.
- Implemented a custom AskT abstraction with Functor, Applicative and Monad instances to compose input workflows, including conditional prompts and optional values.

Budgerigar Weight Monitoring System
- Developed a perch-style weighing device to monitor the weight of a pet budgerigar.
- Built a PC-based data collection and visualisation system to track weight changes over time.
- Awarded the Grand Prize and Special Award in the Electronics Division of the 2019 Miyagi Procon.

AWARDS

- **Grand Prize and Special Award**, Miyagi Programming Contest (Miyagi Procon), 2019.

TECHNICAL SKILLS

- **Languages**: Haskell, C, Python, AVR Assembly
- **Systems & Embedded Development**: Linux (WSL), Win32 API, Bare-metal AVR
- **Tools**: Git, Vim, KiCad, LaTeX

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